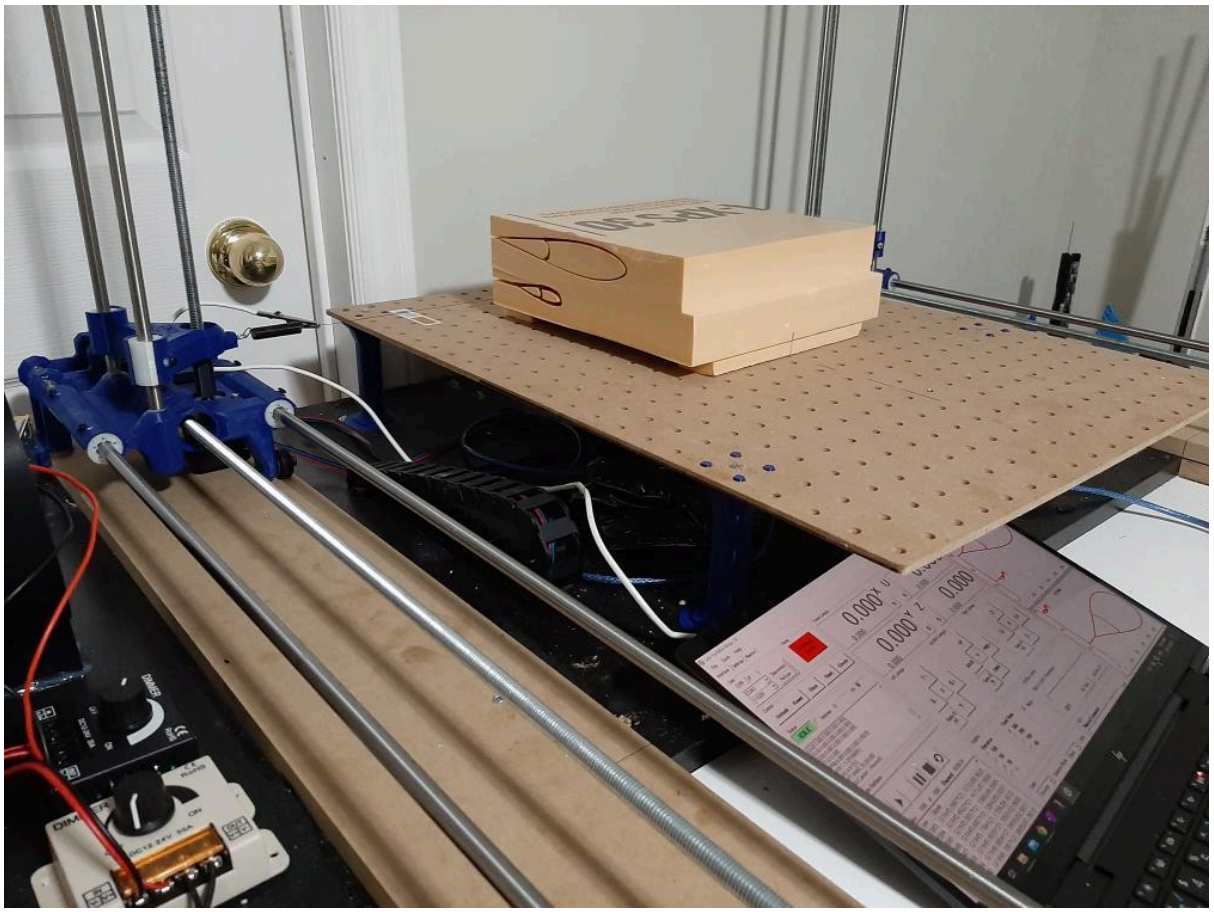
4-Axis Foam Cutting Machine

Objective:
Designed and built a 4-axis CNC foam cutter to create precise aerodynamic parts for planes and cars, used in SAE competitions and personal VTOL projects.



Features:

1. **Machine Design**:
    - Designed the parts of the 4-axis hot-wire foam cutter, including bearings, using **Fusion 360**.
    - Most parts were 3D printed as intended, ensuring cost-effectiveness and precision in assembly.
    - Achieved cutting precision of ±0.5 mm for accurate aerodynamic part production.
2. **Software Integration**:
    - **DevWing Foam**: Used for designing advanced aerodynamic wings.
    - **DevFus Foam**: Utilized for creating precise fuselage designs.
    - **GRBL**: Implemented as the interface software for real-time machine control and G-code execution.
3. **Applications**:
    - Produced high-quality, lightweight parts for SAE car bodies and VTOL wings.
    - Reduced prototyping time by 50% compared to manual cutting processes.

Skills Demonstrated

1. **CNC Machine Design**: Built a functional 4-axis CNC system with parts designed in Fusion 360 and fabricated via 3D printing.
2. **Aerodynamic Wing Design**: Created efficient wings using DevWing Foam.
3. **Fuselage Modeling**: Designed complex fuselage structures with DevFus Foam.
4. **3D Printing**: Successfully fabricated structural and functional components, including bearings, for the CNC machine.
5. **Prototyping**: Delivered performance-optimized parts for aviation and automotive projects.

